ADC System Development Report

R2.1 delivery and next operational stage 系统开发进展与运行保障

Report date 2026-10-18 Application ccdcdcd R2.1 DELIVERED R2 IN PROGRESS

1 Current release and next decision 当前版本与下一步

R2.1 request tracing and safe operational logs（请求追踪与安全运行日志）are delivered on the Mac and in GitHub. Failed operations now have a server reference that connects safe application, server and gateway events. Existing financial approval, clinical permissions and business history remain intact. The next implementation is R2.2 worker and backlog observation.

The agreed product remains one dental chain with approximately 5-20 clinics. Data is synthetic（合成数据）, business currency/time are CNY and Asia/Shanghai, and external AI remains unused. Twenty-clinic capacity, independent hosting and commercial readiness remain separate acceptance targets.

This revision records R2.1 execution on 18 October 2026 at application commit ccdcdcd [11-13]. R1 results dated 15 October 2026 remain historical [3-4]. Future R2 monitoring thresholds, thirty-minute capacity repetition and twenty-four-hour observation have not been accepted by this release.

Area	Current position
Application	R1 financial evidence retained; R2.1 diagnosis/logging delivered
Regression	918 source checks; 1,119 PostgreSQL/HTTP/concurrency tests
Browser	894 isolated assertions; 410 on the updated original Mac
Historical capacity	R1: five clinics, twenty accounts, 30 minutes; 9,000 requests
Recovery	R2.1: 50 independent restore fingerprints; 48 durable Mac fingerprints preserved
Next coding batch	R2.2 worker heartbeat, due backlog and private condition reporting

Cloud acceptance awaits private server and hostname inputs. The original Windows database recovery remains unresolved. The frozen RC1 report remains the historical design record; this report is the current development and planning companion.

Mind map 思维导图 ADC → delivered R1 money evidence + R2.1 safe diagnosis → next R2.2 observation → recovery freshness → independent host and measured expansion.

2 Business workflow and module ownership 业务流程与模块职责

The main patient journey connects booking, arrival, dentist care, actual treatment, the versioned invoice, receipts or refunds and financial review. Equipment repair connects a responsible work order, another manager's acceptance, a service event and separately reviewed expense. Existing appointments and scheduling decisions retain their own protected commands.

Seven workspaces（工作身份）separate function, clinic and record authority. A person with multiple grants activates one role at a time. A role switch does not allow self-review or combine permissions. Clinical records, money and equipment rules remain inside their owning module; callers use defined operations [1-2].

Role	Responsibility and boundary
PATIENT	Own appointments, invoice projection and authorized portal actions
FRONT_DESK	Current-clinic reception, booking and invoice-related receipt/refund evidence
DENTIST	Assigned clinical work and actual treatment; no independent finance approval
MANAGER	Current-clinic equipment, schedule coordination and reconciliation proposals; limited money projection
FINANCE	Channel configuration and independent money/evidence/match review under chain authority
HQ	Authorized aggregate reporting; no detailed statement or patient access from reconciliation
SYSADMIN	Account and directory governance; technical privilege does not grant clinical access

Billed revenue, cash received, equipment costs and approved refunds are distinct facts. Missing cost is distinct from confirmed zero. Suggestions are not appointments, an inbox commit is not human agreement, and an approved daily handover is not proof of external channel settlement. Operational monitoring must preserve each distinction.

Mind map 思维导图 Patient journey → appointments and care; money → invoice, cash and independent review; equipment → repair evidence and explicit scheduling; identity → active role, clinic and object authority.

3 R1 financial evidence closure 资金凭据与逐笔对账

Finance establishes stable clinic-specific channel accounts. New receipt, refund, equipment-expense and reversal requests bind the account and reference into a signed preview. The cash entry and first evidence version are saved atomically; a different finance account decides whether to post. Legacy entries keep an explicit missing-evidence status until a separately reviewed supplement is supplied.

Managers upload private UTF-8 CSV statements with six v1 columns, at most 2 MiB and 1,000 rows. Source bytes, digest, parser version and normalized rows stay in the database. Confirmation uses a server-held preview and reparses the source. Exact repeats are reported and skipped; a conflicting transaction identity rejects the entire acceptance.

Step	Invariant
Propose matching	Same clinic, account, CNY currency, direction, amount and current approved reference
Independent review	Different account; current grants and source facts rechecked under locks
Active correspondence	One statement line to one POSTED cash entry; database uniqueness and transactional rechecks
Correct evidence	Append a reviewed version; invalidate the affected active match and retain its decision
Handle differences	Keep missing evidence, fees, combined deposits and unsupported allocations visible
Daily handover	Keep four gross internal totals separate from current channel matching and next-day settlement

Importing or matching never posts money, changes an invoice or invents a balancing entry. The historical R1 browser scenarios accepted on 15 October cover partial receipts, refunds, equipment payments/reversals and evidence correction. In that R1 run, after additional load-time review, 29 matches are confirmed and one prior decision remains stale; 5,031 statement lines are retained. Those are deliberately bounded synthetic scenarios, not a real reconciliation coverage rate [3].

Mind map 思维导图 Authorized account → signed evidence → retained CSV → exact proposal → different reviewer → readable differences/corrections; internal handover and channel settlement remain separate.

4 Executed evidence and its scope 已执行验证与适用范围

R2.1 separates real database behavior, browser workflows, proxy/command sinks, size rotation and encrypted recovery. Counts describe executed checks, not coverage percentages. The local acceptance and protected Mac delivery records are separate [11-13].

The five-clinic SQL oracle independently retains receipts of CNY 50,100.10, refunds of 100.10 and net patient cash of 50,000.00. Equipment expense and reversal are CNY 50.05 each. Invalid requests, CSV errors and technical logging create no balancing cash or invoice changes.

Check	R2.1 observation
Regression	918 source checks; 1,119 PostgreSQL/HTTP/concurrency tests, including 13 new R2.1 methods
Chrome	568 business + 248 permission/isolation + 78 diagnostic/keyboard/narrow-layout assertions
Real log correlation	28 browser requests match across Django, Gunicorn and Nginx; 63 private marker values absent
Candidate gateway	133 distinct references; login GETs stay outside POST login budget; expected 403/429 cases
Size rotation	250,000 records; 143,591 retained including the final record; production 10m/three-file options
Independent facts	20 SQL/content assertions; 24 confirmed matches, one retained stale decision and 30 accepted lines
Recovery	50 model/association fingerprints match; recorded same-Mac empty-target restore/compare 2.452 s
Mac protection	48 durable fingerprints and 70 private files preserved; same data volume and RC tag; 410 browser assertions

The mixed workload remains R1 historical evidence: five clinics, twenty accounts, 9,000 requests over thirty minutes, p95 0.337 s, under two-CPU/1,728-MiB limits. It was not repeated for R2.1. New restore and gateway checks are local; they do not establish off-host recovery or independent cloud service. Native macOS popup-menu option selection remains outside the earlier automatic acceptance.

Mind map 思维导图 Adversarial inputs → actual three-layer correlation → seven-role business outcomes → independent money facts → size rotation → encrypted restore → exact-commit CI and original Mac preservation.

5 R2 business requirements 运行保障业务需求

R2.1 addresses safe diagnosis when reception, clinical or settlement requests fail: staff receive a support reference and are asked to check stored state before repeating uncertain work. The remaining operational problems are timely detection of stalled coordination notices, an unavailable observer and stale recovery points.

The requirements note makes the expected outcome testable rather than describing the system as merely reliable [1]. Each R2 requirement therefore specifies an actor, an observable condition, a protected invariant, a failure response and an acceptance case.

Requirement	Observable result and protected invariant
R2-01 / 02	Delivered: server reference and strict safe events; no submitted secret, clinical, CSV or URL/body values
R2-03	Delivered size bounds and sink inventory; strict time expiry/protected collection remain gaps
R2-04 / 05	Next: successful worker-loop freshness and eligible due-backlog age; future retry and human read stay distinct
R2-06	Planned: private HEALTHY/WARNING/CRITICAL/UNKNOWN; acknowledgement does not clear the fault
R2-07 / 08	Planned: separate created/downloaded/restored points and controlled independent recovery

Keep the modular monolith（模块化单体）and current worker. Operational logging queries no permissions or business objects and omits role/clinic identifiers. The business module still owns its audit and transaction; operational metadata is a separate diagnostic responsibility [2,5].

Mind map 思维导图 Failure → actor and observable symptom → protected business invariant → explicit response → acceptance; operator, clinic and finance responsibilities stay distinct.

6 Delivered safe diagnosis 已交付的故障定位能力

At request entry the server creates the reference and returns X-Request-ID. Client headers cannot choose it. Context is reset after completion or exceptions; later Django warning records reuse the immutable server context on the request. The gateway uses the application reference for proxied outcomes and generates a fallback for its own refusals.

The JSON formatter allows UTC time, severity, service, identified source or unknown, finite event code, server reference, resolved route, normalized method/status and bounded duration. It never formats arbitrary messages, arguments, exception strings or locals. Authentication events omit entered credentials; references confer no record authority.

Contract	Implementation and check
Context	Separate request module and context variables; original security/workspace middleware order preserved
Safe failures	Generic error and support reference; 500 renderer performs no database context queries
Sinks	Django/Gunicorn/gateway request events controlled; runtime command stderr sanitized; compatible worker aggregate output
Native capture	Database errors can quote submitted values; native DB and credential-bootstrap capture disabled
Business atomicity	Optional log failure preserves success/denial; required financial audit failure still rolls back
Storage	Docker local nominal 10 MiB × 3 files per captured container; size rotation verified, no seven-day deletion claim

Actual checks cover forged references, parallel success/failure, delayed framework logs, login/CSRF, sensitive markers, stale equipment forms and invalid receipts/CSV. Diagnostic metadata, retained financial audit and private operator evidence remain separate [6-8,11]. No business migration, new role, public metrics API or automatic business action was introduced.

Mind map 思维导图 Server reference → existing permission/transaction → strict safe events → matching proxy outcome → bounded private storage → preserved audit and independent review.

7 Observation and recovery targets 监测与恢复目标

These are proposed R2.2/R2.3 engineering defaults, not achieved service results. R2.1 provides diagnostic records and size bounds; it does not yet collect worker/backup conditions or manage incidents. Missing observations must be UNKNOWN. An idle worker should be healthy when its database-backed loop is current and no overdue eligible events exist.

Signal	Initial rule
Web/database readiness	Fault after two failed samples; recovery after two successful samples
Worker success	Stale after 90 seconds when a writable worker is required
Due notification age	Over 120 seconds; use PENDING rows already due, not future scheduled retry
Observer freshness	Over 90 seconds, invalid clock or foreign instance means UNKNOWN
OOM/restart	Any new event retains an incident even when current health recovers
Storage	Warn below 20% free; critical below 10% or 1 GiB free
Recovery points	Warn after 26 hours for daily server backup or verified off-host source point
Independent restore	After deployment and weekly; age warning after eight days; bind backup digest and commit

Track backup creation, verified Mac download and independent restore separately. A repeated check/download never advances the original backup timestamp. Keep seven server days and thirty Mac days; a corrupt file must not cause pruning of a valid recovery point. A sleeping Mac leaves the off-host recovery point at its last verified download.

Recover only into a distinct empty target with matching source/image/schema provenance. Compare accounts, grants, business/association rows, all four cash legs, evidence versions, original CSV byte digests and match history. Target thirty minutes after the replacement environment is ready. R2.1's 2.452-second local interval does not promise cloud replacement time.

A proposed seven-day technical availability objective is 99.5%; report 4xx/409/429 separately. Functional acceptance still rejects an unexpected 403 or other wrong business outcome. Empty traffic is not 100% availability. Repeat the existing thirty-minute mixed-load gate with p95 within five seconds and measure observer/logging overhead.

Mind map 思维导图 Delivered diagnosis → proposed worker/backlog/observer freshness → explicit future condition; created → downloaded/decrypted → independently restored point → actual recovery exposure.

8 Development sequence and completion gates 开发顺序与验收

R2.1 is delivered with passing application-commit CI, local execution and protected Mac comparison. Continue with bounded R2.2 increments. Each must preserve original data, role scope and independent finance review. Destructive fault injection stays inside an identified disposable rehearsal.

Increment	Deliverable and completion gate
R2.1 Safe diagnosis	Delivered: references, strict events, safe errors, measured rotation and protected local release
R2.2 Worker observation	Next: heartbeat, due age, private collector and condition/incident lifecycle; expose observer failure
R2.3 Recovery freshness	Separate backup/download/restore receipts; corrupt/wrong-key/stale-copy and empty-target recovery cases
R2.4 Local release	Security gap review, fault runbooks, seven-role browser/load checks and 24-hour local observation
R2.5 Cloud release	Independent HTTPS host; mainland access, Mac offline, restart, renewal, backup/recovery and 24-hour gates

- Inject database unavailability, stopped/terminated worker, future retry, monitor failure and simulated storage pressure without modifying the running Mac.
- Verify deduplicated incident opening, acknowledgement and sustained recovery; acknowledgement changes responsibility, not condition truth.
- Check corrupt archives, cached copies with a wrong key and interrupted downloads; preserve previous valid points and all source business records.
- Repeat five-clinic/twenty-account valid mixed work with import/review activity, no unexpected statuses/duplicate effects/OOM and p95 within five seconds.

The full plan defines R2-A01 through R2-A12 [9]. R2.1 contributes reference/privacy/permission, storage and recovery evidence; it does not pass every full-R2 gate. Local R2 acceptance and cloud acceptance remain separate. Then retain the agreed sequence: R3 treatment continuity, R4 resource coordination and R5 measured twenty-clinic expansion.

Mind map 思维导图 Diagnosis → worker/observer → recoverable points → local fault evidence → independent host; later care continuity → shared resources → measured expansion.

9 Security assessment and release boundaries 安全差距与发布边界

Use OWASP ASVS 5.0.0 for the scoped assessment. R2.1 request/context, safe-sink and retained-business tests now support evidence review; they do not establish a standard-wide pass. The register keeps time retention, protected independent collection, session/MFA and target-host work explicit [6,10-11].

Area	Current assessment direction
Logging and errors	Implemented R2.1 inventory, correlation, UTC/minimal fields, safe failures and adversarial evidence; assess complete controls
Log protection	Size rotation verified; seven-day deletion and separate protected collection remain gaps; native text suppression reduces forensic detail
Authorization	Review existing seven-role/function/object/field tests against the complete requirement scope
Sessions and MFA	Lifetime/concurrency, session inspection, factor changes and MFA need their own contracts/assessment
Customer isolation	One chain’s clinic isolation is not independent customer tenancy
Public operation	Needs server IPv4, SSH user/key path, independently verified host fingerprint and owned hostname

The application is updated on the original Mac after exact-commit CI. Encrypted before/after backups are verified; original durable facts, configuration, volume and RC tag remain. Documentation is synchronized separately. No test business records are injected into the original Mac.

Local checks cannot establish mainland access, real certificate issuance/renewal, service while the Mac is offline or replacement-host recovery. Cloud acceptance awaits server details. Original Windows-data recovery remains unresolved.

Mind map 思维导图 Versioned security requirement → existing evidence and explicit gap → targeted work; private server inputs → actual host tests → separately accepted public operation.

10 References and report provenance 参考资料与报告来源

The course notes provide the requirements/design framework; official references guide engineering mechanisms. R1 observations remain dated 15 October 2026. R2.1 local acceptance and original-Mac delivery are dated 18 October 2026, with identified application commit ccdcdcd and passing CI [11-13].

Editable report_source.json and build_report.py produce Word and Markdown; PDF pages are visually inspected. baseline.json binds aggregate evidence and delivery hashes. Credentials, keys, dumps, patient data and CSV source bytes remain private. The frozen 63-page RC1 report, original thirty migrations and RC tag preserve their historical identities.

Report revision date: 2026-10-18 in the project owner's client date context. Historical R1 evidence remains dated 2026-10-15 UTC. The separate R2.1 delivery record binds 48 durable-model/association fingerprints, 70 unchanged private files and 410 Mac browser assertions to this application release.

[1] Course note 3 Requirements and specification. Supplied project material, physical page 5 on testable observable requirements.

[2] Course note 5 Software design principles. Supplied project material, physical pages 4-5 and 8 on separation, simplicity, restricted visibility and information hiding.

[3] ADC R1 executed acceptance. Immutable application baseline; synthetic local scope.

[4] Exact R1 application commit CI. Successful workflow for commit 6a06f35.

[5] Google SRE workbook Monitoring. Purposeful observation and testing the monitoring/alerting path.

[6] OWASP ASVS official project and stable standard. Assessment is pinned to 5.0.0; requirement links are in the ADC register.

[7] Django 5.2 logging configuration. Application logger, formatter and handler mechanisms.

[8] Docker local logging driver. Size-based options; no time-expiry acceptance is claimed.

[9] ADC R2 operational assurance plan. Implementation contracts and R2-A01 through R2-A12; planned.

[10] ADC versioned security gap register. Scoped assessment work, not ASVS certification.

[11] ADC R2.1 executed local acceptance. Regression, real-browser/sink/gateway, money and encrypted-restore results.

[12] Exact R2.1 application CI. Successful workflow for ccdcdcd.

[13] ADC protected Mac delivery record. Original data/configuration protection and post-update verification.

Mind map 思维导图 Course requirements/design → official mechanisms → immutable R1 observations → executed R2.1 acceptance → original Mac delivery → pending R2.2 and independent host.